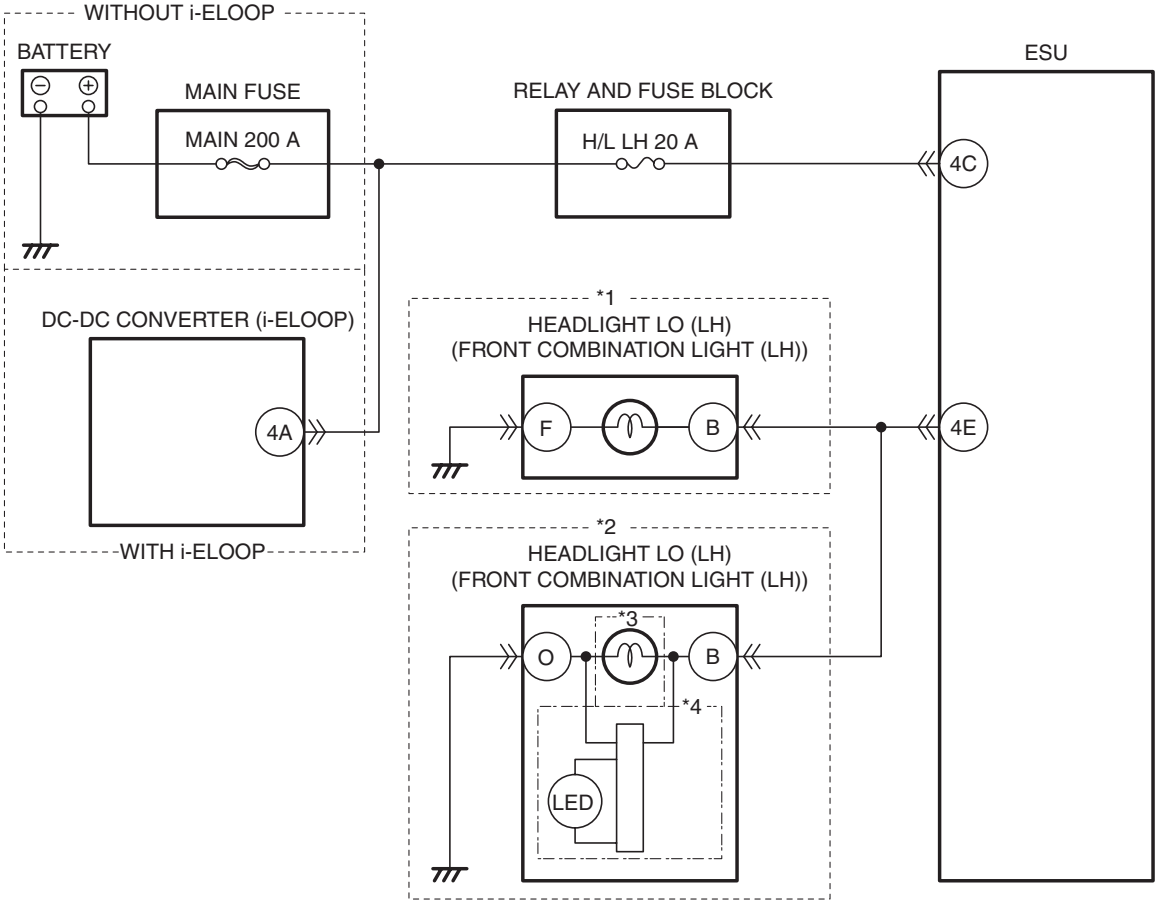
## DTC B1D00:13 [ELECTRICAL SUPPLY UNIT (ESU)]

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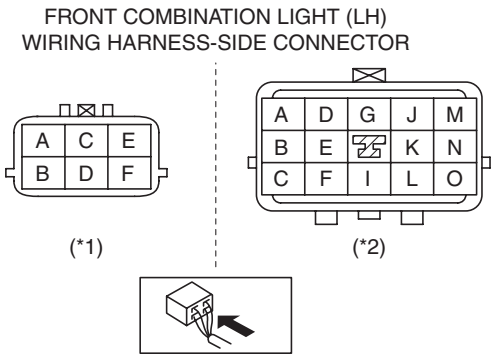
| Description | Headlight LO (LH) circuit malfunction |
|---|---|
| Detection condition | • Electrical supply unit (ESU) detects an open circuit in the headlight LO (LH) circuit. |
| Fail-safe | Not applicable |
| Possible cause | • Electrical supply unit (ESU) connector or terminal malfunction<br>• Electrical supply unit (ESU) power supply circuit malfunction<br>— Without i-ELOOP:<br>• Short to ground in wiring harness between MAIN 200 A fuse and electrical supply unit (ESU) terminal 4C<br>• H/L LH 20 A fuse malfunction<br>• MAIN 200 A fuse malfunction<br>• Open circuit in wiring harness between battery positive terminal and electrical supply unit (ESU) terminal 4C<br>— With i-ELOOP:<br>• Short to ground in wiring harness between H/L LH 20 A fuse and electrical supply unit (ESU) terminal 4C<br>• H/L LH 20 A fuse malfunction<br>• Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and electrical supply unit (ESU) terminal 4C<br>• Headlight HI/LO bulb (LH) malfunction<br>• Front combination light (LH) connector or terminal malfunction<br>• Open circuit in wiring harness between the following terminals:<br>— Front combination light (LH) terminal F—Body ground (With front combination light (bulb type) and without headlight manual leveling system)<br>— Front combination light (LH) terminal O—Body ground (With front combination light (LED type) or headlight manual leveling system)<br>• Open circuit in wiring harness between front combination light (LH) terminal B and electrical supply unit (ESU) terminal 4E<br>• Front combination light (LH) malfunction<br>• Electrical supply unit (ESU) malfunction |

| Description | Headlight LO (LH) circuit malfunction |
|---|---|










*1 : With front combination light (bulb type) and without headlight manual leveling system

*2 : With front combination light (LED type) or headlight manual leveling system

*3 : With front combination light (bulb type)

*4 : With front combination light (LED type)

**Diagnostic Procedure**

| Step | Inspection | | Action |
|---|---|---|---|
| 1 | **INSPECT ELECTRICAL SUPPLY UNIT (ESU) CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Disconnect the electrical supply unit (ESU) connector.<br>• Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection.<br>• Is the connector normal? | Yes | **Without i-ELOOP:**<br>• Go to the next step.<br>**With i-ELOOP:**<br>• Go to Step 3. |
| | | No | Repair or replace the connector, then go to Step 9. |
| 2 | **VERIFY ELECTRICAL SUPPLY UNIT (ESU) POWER SUPPLY VOLTAGE**<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Measure the voltage at the electrical supply unit (ESU) terminal 4C (wiring harness-side).<br>• Is the voltage **B+**? | Yes | **With front combination light (bulb type):**<br>• Go to Step 4.<br>**With front combination light (LED type):**<br>• Go to Step 5. |
| | | No | Inspect the H/L LH 20 A fuse and MAIN 200 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and electrical supply unit (ESU) terminal 4C.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>• Replace the fuse.<br>• If the fuse is damaged:<br>— Replace the fuse.<br>• If the fuse is normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and electrical supply unit (ESU) terminal 4C.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 9. |

| Step | Inspection | | Action |
|---|---|---|---|
| 3 | **VERIFY ELECTRICAL SUPPLY UNIT (ESU) POWER SUPPLY VOLTAGE**<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Measure the voltage at the electrical supply unit (ESU) terminal 4C (wiring harness-side).<br>• Is the voltage **B+**? | Yes | **With front combination light (bulb type):**<br>• Go to the next step.<br>**With front combination light (LED type):**<br>• Go to Step 5. |
| | | No | Disconnect the service plug.<br>(See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)<br>Inspect the H/L LH 20 A fuse.<br>• If the fuse is blown:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between H/L LH 20 A fuse and electrical supply unit (ESU) terminal 4C.<br>**If there is a common connector:**<br>• Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness.<br>• Replace the fuse.<br>• If the fuse is damaged:<br>— Replace the fuse.<br>• If the fuse is normal:<br>— Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and electrical supply unit (ESU) terminal 4C.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Connect the service plug, then go to Step 9.<br>(See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) |
| 4 | **VISUALLY INSPECT HEADLIGHT HI/LO BULB (LH)**<br>• Visually inspect the headlight HI/LO bulb (LH).<br>• Is the headlight HI/LO bulb (LH) normal? | Yes | Go to the next step. |
| | | No | Replace the headlight HI/LO bulb (LH), then go to Step 9.<br>(See HEADLIGHT BULB REMOVAL/INSTALLATION.) |
| 5 | **INSPECT FRONT COMBINATION LIGHT (LH) CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Disconnect the front combination light (LH) connector.<br>• Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection.<br>• Is the connector normal? | Yes | Go to the next step. |
| | | No | Repair or replace the connector, then go to Step 9. |

| Step | Inspection | | Action |
|---|---|---|---|
| 6 | **INSPECT HEADLIGHT LO (LH) GROUND CIRCUIT FOR OPEN CIRCUIT**<br>• Verify that the front combination light (LH) connector is disconnected.<br>• Inspect for continuity between the following terminals (wiring harness-side) and body ground:<br>— Front combination light (LH) terminal F (With front combination light (bulb type) and without headlight manual leveling system)<br>— Front combination light (LH) terminal O (With front combination light (LED type) or headlight manual leveling system)<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• Front combination light (LH) terminal F—Body ground (With front combination light (bulb type) and without headlight manual leveling system)<br>• Front combination light (LH) terminal O—Body ground (With front combination light (LED type) or headlight manual leveling system)<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 9. |
| 7 | **INSPECT HEADLIGHT LO (LH) CIRCUIT FOR OPEN CIRCUIT**<br>• Verify that the front combination light (LH) and electrical supply unit (ESU) connectors are disconnected.<br>• Inspect the wiring harness for continuity between front combination light (LH) terminal B (wiring harness-side) and electrical supply unit (ESU) terminal 4E (wiring harness-side).<br>• Is there continuity? | Yes | **With front combination light (bulb type):**<br>• Go to Step 9.<br>**With front combination light (LED type):**<br>• Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between front combination light (LH) terminal B and electrical supply unit (ESU) terminal 4E.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 9. |
| 8 | **VERIFY IF MALFUNCTIONING LOCATION IS FRONT COMBINATION LIGHT (LH) DEPENDING ON REPEATABILITY**<br>• Always reconnect all disconnected connectors.<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].)<br>• Operate the light switch to the HEAD position and the dimmer switch to LO position.<br>• Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].)<br>• Is the same DTC displayed? | Yes | Replace the front combination light (LH), then go to the next step.<br>(See FRONT COMBINATION LIGHT REMOVAL/ INSTALLATION.) |
| | | No | Go to Step 10. |

| Step | Inspection | | Action |
|---|---|---|---|
| 9 | **VERIFY THAT REPAIRS HAVE BEEN COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)<br>• Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].)<br>• Operate the light switch to the HEAD position and the dimmer switch to LO position.<br>• Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].)<br>• Is the same DTC displayed? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the electrical supply unit (ESU). (See ELECTRICAL SUPPLY UNIT (ESU) REMOVAL/INSTALLATION.)<br>Go to the next step. |
| | | No | Go to the next step. |
| 10 | **VERIFY IF OTHER DTCs DISPLAYED**<br>• Are any other DTCs displayed? | Yes | Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ELECTRICAL SUPPLY UNIT (ESU)].) |
| | | No | DTC troubleshooting completed. |